Quasinormal Modes of Schwarzschild Black Holes

Numerical Methods for Differential Equations

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1 Introduction

The merger of two black holes (BHs) can be divided into three main phases: the inspiral, during which the two black holes orbit around each other; the merger, during which they coalesce; and the ringdown, during which the newly formed black hole relaxes towards a stationary configuration. Gravitational waves (GWs) are emitted throughout all three phases, as illustrated in Fig. 1.

During the ringdown, the remnant black hole emits gravitational waves at characteristic complex frequencies known as quasinormal modes (QNMs). This process is analogous to the sound produced by a bell after it has been struck: the signal can be decomposed into characteristic oscillation frequencies, each with its own damping time. In the black-hole case, these frequencies are determined by the properties of the black hole and by the equations governing perturbations of spacetime.

The inspiral phase has so far provided particularly strong constraints on the parameters of observed binary black holes. However, the ringdown offers a complementary way to probe the properties of the remnant. Measuring several QNMs could allow us to infer its mass and angular momentum independently and to test whether the remnant is consistent with the predictions of general relativity.

The aim of this project is to calculate black-hole QNMs numerically and investigate the numerical challenges associated with this problem. As a first step, we consider a non-rotating, uncharged black hole, described by the Schwarzschild solution. This simplified setting already leads to a non-trivial eigenvalue problem with complex eigenfrequencies and radiative boundary conditions.

2 A Brief Introduction to Quasinormal Modes

2.1 The perturbation equation

The ringdown can be studied by considering small perturbations of the geometry of spacetime. For a Schwarzschild black hole, the linearized gravitational per-

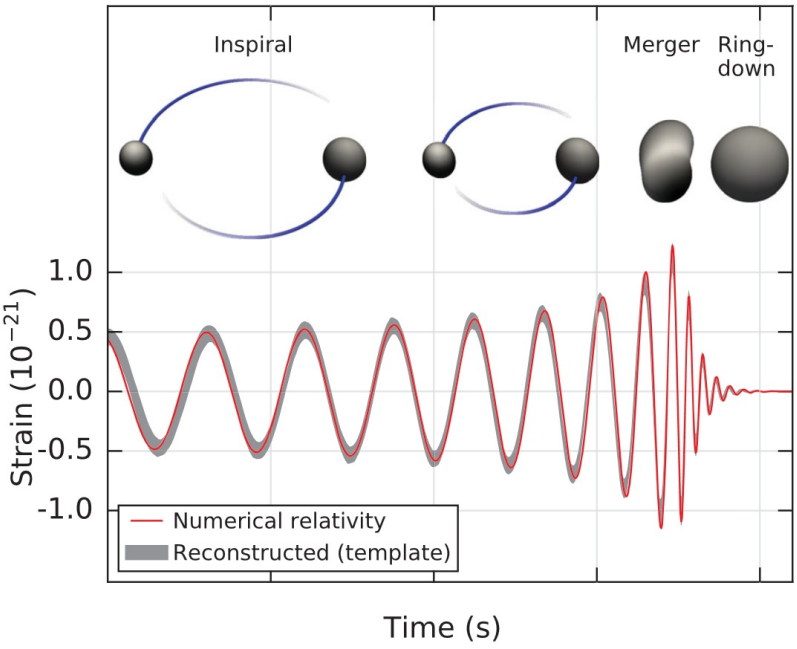


Figure 1: Gravitational waves emitted during a binary black hole merger [1].

turbation equations can be reduced to a one-dimensional wave equation, known as the Regge–Wheeler equation.

Using geometrized units $G = c = 1$, the Regge–Wheeler equation for a gravitational perturbation can be written as

$$\frac{\partial^2 \Psi}{\partial t^2} - \frac{\partial^2 \Psi}{\partial r_*^2} + V_\ell(r)\Psi = 0, \tag{1}$$

where Ψ is the perturbation variable, ℓ is the spherical-harmonic multipole index, and r_* is the tortoise coordinate, defined by

$$r_* = r + 2M \ln\left(\frac{r}{2M} - 1\right). \tag{2}$$

Here, M is the black-hole mass and $r > 2M$ is the Schwarzschild radial coordinate outside the event horizon.

The effective Regge–Wheeler potential for gravitational perturbations is

$$V_\ell(r) = \left(1 - \frac{2M}{r}\right)\left[\frac{\ell(\ell+1)}{r^2} - \frac{6M}{r^3}\right], \qquad \ell \geq 2. \tag{3}$$

The potential describes the scattering of perturbations by the spacetime curvature outside the black hole. It vanishes both at the event horizon and at spatial infinity.

Seeking solutions with harmonic time dependence,

$$\Psi(t, r) = e^{-i\omega t}\psi(r), \tag{4}$$

we obtain the radial equation

$$\frac{d^2\psi}{dr_*^2} + \left[\omega^2 - V_\ell(r)\right]\psi = 0, \tag{5}$$

where ω is, in general, a complex frequency.

2.2 Quasinormal-mode boundary conditions

For an ordinary vibrating string, the allowed normal-mode frequencies are determined by the boundary conditions at its endpoints. A black hole presents a different problem: waves can propagate into the event horizon or escape to infinity. Quasinormal modes are therefore defined by purely ingoing boundary conditions at the horizon and purely outgoing boundary conditions at spatial infinity.

With the time dependence $e^{-i\omega t}$, these conditions take the asymptotic form

$$\psi(r_*) \sim e^{-i\omega r_*}, \qquad r_* \to -\infty, \tag{6}$$

$$\psi(r_*) \sim e^{+i\omega r_*}, \qquad r_* \to +\infty. \tag{7}$$

The first condition describes a wave entering the future event horizon, while the second describes a wave propagating outward towards infinity.

These boundary conditions select a discrete set of complex frequencies, $\omega_{\ell n}$, where ℓ is the angular multipole index and n is the overtone number. With our time-dependence convention, the real part of $\omega_{\ell n}$ sets the oscillation frequency, while a negative imaginary part describes exponential damping.

For a mode with $\omega = \omega_R + i\omega_I$, the oscillation frequency and damping time are

$$f = \frac{\omega_R}{2\pi}, \qquad \tau = -\frac{1}{\omega_I}, \qquad \omega_I < 0. \tag{8}$$

The QNM problem is thus a complex eigenvalue problem with non-standard boundary conditions. It is also numerically challenging: for damped modes, the radial solutions satisfying the QNM boundary conditions can grow exponentially in one direction. Consequently, straightforward numerical shooting can become unstable, and small numerical errors may excite unwanted components of the solution.

A more detailed introduction to QNMs can be found in the review [3].

3 Possible Approaches to the Project

The goal is to compute one or more QNM frequencies of a Schwarzschild black hole and assess the accuracy and reliability of the numerical method. Several approaches are possible, depending on the time available and the desired level of difficulty.

1. **Which perturbation equation will you solve?** A natural starting point is the Regge–Wheeler equation for axial gravitational perturbations. An extension would be to solve the Zerilli equation [3] and compare the resulting frequencies. More advanced projects could investigate the Teukolsky equation for rotating Kerr black holes.

2. **Which numerical method will you use?** Possible approaches include shooting methods, frequency-domain spectral methods, and time-domain evolution followed by a fit to the ringdown signal. Pseudospectral methods are one particularly interesting option; see, for example, [2]. Compare the advantages and limitations of the methods you implement.

3. **How will you impose the boundary conditions?** The QNM boundary conditions are imposed at the horizon and at spatial infinity, which correspond to $r_* \to -\infty$ and $r_* \to +\infty$. In practice, the computational domain must be handled using finite boundaries, coordinate transformations, asymptotic expansions, or another suitable technique. How does your choice affect the accuracy and stability of the calculation?

4. **How will you validate your results?** Compare your computed frequencies with published Schwarzschild QNM tables or with suitable approximations, such as the Wentzel–Kramers–Brillouin (WKB) method. Investigate convergence with respect to the grid resolution, spectral truncation order, or integration tolerances. If possible, compare two independent numerical methods.

5. **How will you identify numerical instabilities?** A numerical solution can contain unwanted components that do not satisfy the intended boundary conditions. Study the sensitivity of the computed frequencies to initial guesses, domain size, resolution, and numerical precision. How can you distinguish a genuine QNM from a numerical artifact?

6. **What physical properties will you investigate?** Study how the QNM frequencies depend on the multipole index ℓ and overtone number n. Verify the expected scaling with black-hole mass and investigate the relationship between oscillation frequency and damping time.

A successful project need not implement a state-of-the-art QNM solver. Computing a small number of modes accurately, understanding the numerical difficulties, and demonstrating convergence can already provide valuable insight into complex eigenvalue problems.

References

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